

PRESS

A Player-Level Forechecking Metric

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The Problem with Measuring Forechecking

- A strong forecheck creates turnovers and scoring chances – but players who drive it often get no credit
- Traditional stats capture who ends up with the puck, not who forced the mistake
- Off-puck actions — cutting off passing lanes, applying indirect pressure — go uncredited
- No publicly available metric for measuring individual forechecking contributions

Introducing PRESS

- Puck Recovery and Exit Suppression Score
- Measures how much each player's actions impact probability of a forecheck creating a turnover
- Captures off-puck contributions — closing gaps, disrupting passing lanes— not just takeaways

SETUP

Value created by how a player positions themselves at the start of the forecheck

EXECUTION

Value created by how a player positions themselves as the forecheck unfolds

Building PRESS

28,000+

forechecks

~36%

average forecheck
success rate

Successful forecheck = forechecking team recovers the puck before it leaves the zone

1 Team Credit

Determine how much credit the team deserves for the forecheck outcome

2 Decomposition

Split that credit into the Setup and Execution components of PRESS

3 Individual Attribution

Allocate credit to each player based on their contribution vs. an average forechecker

Step 1: Assigning Team Credit

Baseline success rate across 28,000+ forechecks:

36%

Average forecheck
success rate

✓ **Forecheck Succeeds**

Team Credit = $1 - 0.36 = +0.64$

Positive credit is given for an above average outcome

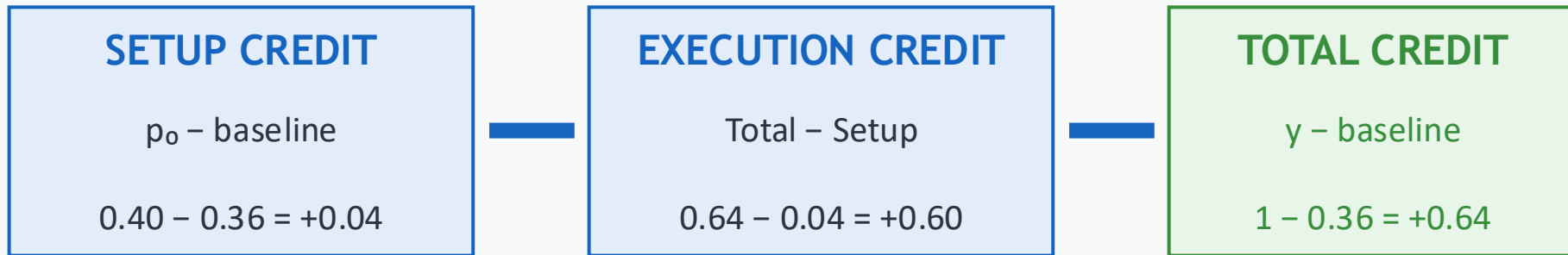
✗ **Forecheck Fails**

Team Credit = $0 - 0.36 = -0.36$

Negative credit is given for a below average outcome

Step 2: Decomposing Team Credit

Example: a successful forecheck where the opening snapshot predicts 40% eventual recovery probability

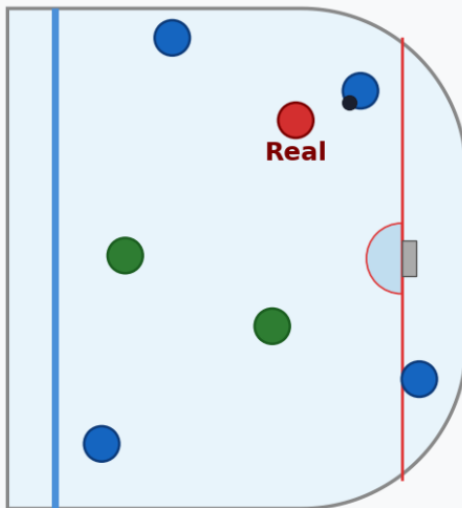


Setup captures positioning value at start of the forecheck. Execution captures positioning value for everything after.

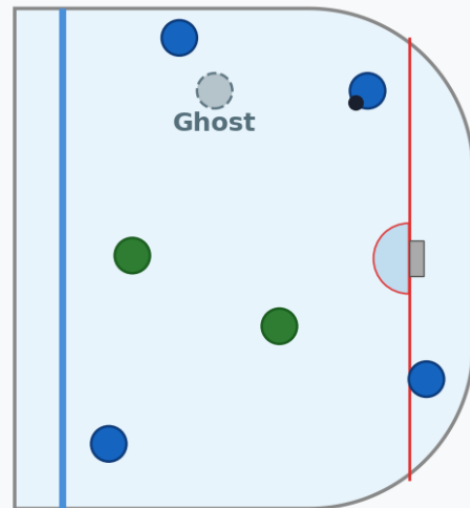
Step 3: Individual Credit Using Ghosts

- We compare each player's positioning and movement to what a hypothetical, average forechecker ("ghost") would have done
- Ghosts are drawn from similar situations in our data
- Estimate probability of a turnover using real player, then with the ghost
- Difference tells us how much the real player, compared to an average forechecker, impacted the outcome
- Each player's credit is allocated based on how much they outperformed or underperformed their ghost

Real Configuration



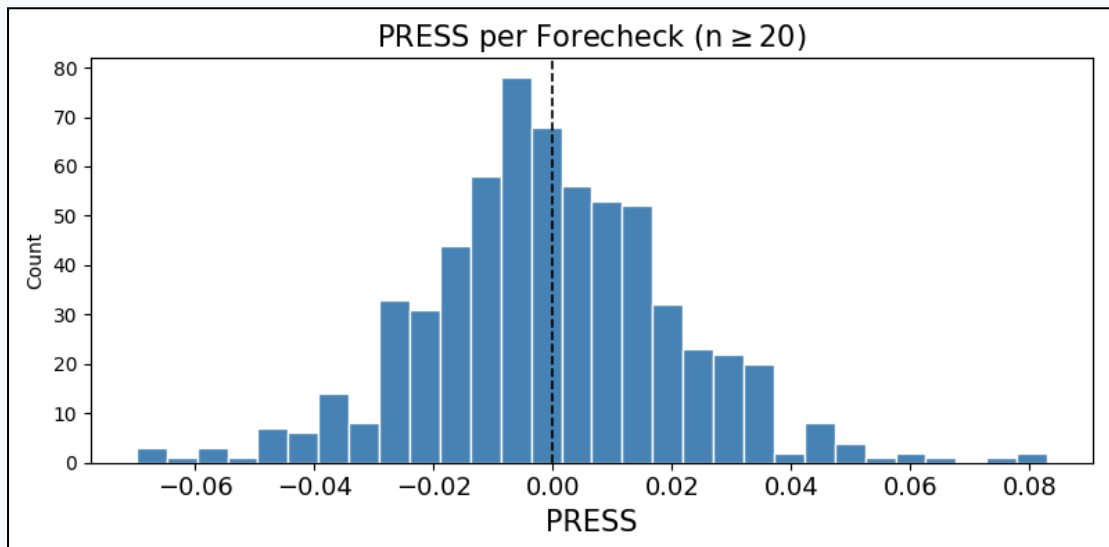
Ghost Comparison



$$\text{Player Impact} = P(\text{Turnover} \mid \text{Real}) - P(\text{Turnover} \mid \text{Ghost})$$

The PRESS Value

- PRESS = average credit per forecheck across all forechecks a player participated in
- Positive PRESS means the player increased turnover probability above an average forechecker
- Negative PRESS means the player decreased turnover probability compared to an average forechecker



Leaderboard

values are per-forecheck, players must have at least 20 total forechecks

Rank	Player	Pos	Total Forechecks	Setup	Exec	PRESS
1	Anderson, Joey	RW	35	0.030	0.053	0.083
2	Lapierre, Hendrix	C	42	0.012	0.070	0.082
3	Wedman, Matthew	LW	70	-0.006	0.081	0.075
4	Firstov, Vladislav	F	27	0.016	0.050	0.066
5	Kiviranta, Joel	LW	21	0.021	0.040	0.060
6	Bitten, William	F	183	0.003	0.050	0.053
7	Vecchione, Mike	F	176	-0.013	0.064	0.050
8	Shaw, Mason	LW	59	-0.002	0.051	0.048
9	Fortier, Gabriel	C	143	-0.035	0.083	0.048
10	Carpenter, Ryan	C	33	-0.001	0.048	0.047

PRESS Outperforms Simpler Baselines

- Validated PRESS against two baseline credit methods
- Baseline 1 — Participation: equal credit to all players on ice at forecheck end
- Baseline 2 — Distance: credit inversely proportional to distance from puck carrier
- Tested how well each method predicted forecheck outcomes (success or failure)

Credit Attribution Method	Log Loss
PRESS	0.64960
Baseline 2 - Distance	0.64997
Baseline 1 - Participation	0.65006

Practical Applications

Scouting

Find undervalued
forecheckers whose impact
is invisible in box scores

Player Development

Setup vs. Execution
breakdown pinpoints where
a player's deficiencies are—
enabling targeted coaching

In-Game Decisions

Match personnel to
forecheck-heavy game
plans

Closing

- PRESS provides a way to compare forechecking skill between players
- Combines tracking data and ghost comparisons to measure how much each individual forechecker contributes to creating turnovers
- Breaks down skill into Setup and Execution

Thank you